LITERATURE SURVEY REPORT

Topic: Skillora AI: AI-Powered Career Readiness & Skill Gap Analyzer

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1. Introduction & Project Context

In the current competitive employment landscape, students often struggle to understand whether their academic training matches practical industry expectations. Traditional career preparation typically relies on static resume guidelines, high-level ATS keyword checks, and manual exploratory research of job profiles. This fragmented approach leaves critical questions unanswered: 'What exact competencies am I missing for my dream role?', 'What should I learn next?', and 'How ready am I to apply?'.

To resolve these issues, the proposed project, Skillora AI, serves as an intelligent platform going beyond basic text scanning. It extracts information dynamically from a student's unstructured resume (PDF/DOCX), maps it directly to targeted job domain frameworks (e.g., Data Analyst, ML Engineer, Software Developer), identifies precise technical or behavioral gaps, ranks them according to critical modern workflows, and offers automated structured upskilling recommendations. This literature survey consolidates foundational research methodologies in text mining, semantic intelligence, transformer-based encoders, and recommendation loops that validate the structural framework of Skillora AI.

2. Comprehensive Literature Review

Paper 1: AI-Driven Semantic Matching and Skill Gap Analysis Using Sentence-BERT

Source: Design and Implementation of an AI-Driven Skill Gap Analysis System (JOIREM, 2026).

Objectives: To automate the cross-referencing process between candidate capabilities and highly dynamic job descriptions without losing context through strict keyword dependencies.

Methodology: The framework transitions away from traditional text-filtering filters by adopting a deep semantic similarity technique. It leverages Sentence-BERT (SBERT) fine-tuned on the all-MiniLM-L6-v2 architecture to convert resume sentences and job demands into unified high-dimensional vector spaces, calculating precise cosine similarity indexes. An isolated analysis layer handles multi-role matching and exposes the spatial metrics on a visual 3D dashboard.

Key Findings: The system successfully eliminated keyword rigidities (e.g., recognizing that 'Statistical Modeling' matches semantic variants like 'Regression Analysis'). It successfully quantified explicit competency gaps simultaneously across multiple target configurations, providing personalized directional milestones.

Research Gaps: The architecture does not handle deeply layered structural sub-dependencies within complex task descriptions or evaluate the explicit real-world authority of the matching projects.

Future Scope: Integration of temporal weights to factor in skill decay and expanding vector capabilities to support real-time market data streaming.

Paper 2: AI-Based Smart Resume Parser and Skill Extractor Using Bi-LSTM-CRF and Keyword Matching

Source: AI-Based Smart Resume Parser and Skill Extractor (ResearchGate, 2026).

Objectives: To structure information from highly unstructured, variably formatted multi-style professional profiles with high lexical accuracy.

Methodology: The researchers implemented a unified hybrid Natural Language Processing (NLP) framework. It combines a fast dictionary-based keyword look-up rule mechanism with a deep sequential deep learning framework built on Bidirectional Long Short-Term Memory joined with Conditional Random Fields (Bi-LSTM-CRF) for advanced Named Entity Recognition (NER). A specialized fusion controller aggregates outputs from both tracks to filter anomalies.

Key Findings: The combined hybrid network achieved an overall entity extraction validation accuracy of 91.6%, vastly outstripping independent keyword matching (83.2%) and pure unconstrained Bi-LSTM-CRF implementations (89.1%) by enforcing domain-specific constraints.

Research Gaps: The framework is heavily reliant on structured English language patterns and exhibits performance bottlenecks when parsing highly complex multi-column visual document structures.

Future Scope: Enhancing the text abstraction stack to support native multi-lingual extraction and integration with modern computer vision layout tools like LayoutLM.

Paper 3: Automated Resume Parsing and Job Domain Prediction Using Machine Learning

Source: Automated Resume Parsing and Job Domain Prediction (2026).

Objectives: To parse, normalize, and automatically classify candidate historical data into exact specialized operational clusters while explaining model criteria.

Methodology: Raw multi-format profiles are ingested and normalized utilizing custom Python regular expressions combined with spaCy's pretrained spatial pipelines. Once segmented into discrete functional profiles (Education, Projects, Technical Core), an ensemble machine learning stack powered by CatBoost models and K-Nearest Neighbors (KNN) evaluates alignment profiles.

Key Findings: The KNN classification framework recorded a peak accuracy of 92.5% for structural job category mapping. The integration of advanced model interpretability helpers provided clear feature-weight data showing why individual records mapped to designated roles.

Research Gaps: The system is restricted to a set of static dataset categories and does not provide an actionable sequential step-by-step roadmap to eliminate identified professional weaknesses.

Future Scope: Transitioning to a dynamic training topology that updates its target classes automatically from active continuous job board indices.

Paper 4: AI-Powered Reskilling Pathways for Closing Digital Skill Gaps

Source: AI-Powered Reskilling Pathways Using SAP Talent Intelligence Hub (ResearchGate, 2025).

Objectives: To engineer scalable automated enterprise architectures that close identified workspace skill deficiencies through adaptive educational paths.

Methodology: The system orchestrates a multi-step analytical loop utilizing qualitative and quantitative layers. It maps current enterprise skill logs against targeted digital states mapped out by strategic corporate goals. When gaps are isolated, an automated ingestion pipeline designs course syllabi mapping directly to Bloom's Taxonomy scales.

Key Findings: The architecture effectively customized educational workflows to match unique employee proficiency strata, moving past uniform training schedules to generate optimal knowledge gains per credit hour invested.

Research Gaps: The framework operates effectively within highly integrated corporate ecosystems but lacks open-source modular components necessary for direct individual student deployments.

Future Scope: Integrating decentralized open learning modules (such as YouTube, GitHub, open MOOCs) into the generation pipeline to lower training infrastructure costs.

3. Literature Survey Summary Matrix

Paper Topic & Source	Core Objectives	Methodology & Architecture	Key Findings & Achievements	Identified Research Gaps

Semantic Analysis (JOIREM, 2026)	Automate resume-to-job matching with contextual semantic tracking.	Sentence-BERT (SBERT) and all-MiniLM-L6-v2 vector embeddings + 3D visualization.	Overcame strict keyword limitations. Found hidden semantic relations.	Lacks sub-dependency checking within long descriptions.
Hybrid Resume Parsing (ResearchGate, 2026)	Accurate entity extraction from multiple formatting styles.	Hybrid system joining Keyword Matching with Bi-LSTM-CRF frameworks.	Attained 91.6% accuracy. Enforced domain rules over loose model predictions.	Fails on unconventional multi-column graphical documents.
Domain Categorization (Research, 2026)	Classify candidate profiles into career domains with explanations.	spaCy NLP parsing pipeline + CatBoost and KNN Ensemble classifiers.	Achieved 92.5% classification accuracy with explicit feature-weight metrics.	Static classifications; no automated learning roadmap loop.
Reskilling Pathways (SAP, 2025)	Create adaptive learning paths based on workforce gap logs.	Multi-modal competency matrices mapped directly to Bloom's Taxonomy frameworks.	Delivered personalized training roadmaps matching exact proficiency tiers.	Tightly coupled to corporate systems; lacks open-source ingestion.

4. Research Gaps & Architectural Alignment with Skillora AI

The comprehensive literature survey reveals clear developmental hurdles that Skillora AI aims to close:

1. Bridge the Prep-Gap: Most current applications focus strictly on either the employer's screening process or basic keyword matching, skipping the crucial preparation step of 'What should I improve before applying?'. Skillora AI shifts this target focus directly to the student.

2. Layout & Formatting Resilience: While deep learning networks provide high accuracy (such as the 91.6% parsed via Bi-LSTM-CRF frameworks), they often struggle with non-standard visual resume structures. Skillora AI introduces an extraction system that blends robust structural normalization with structural fallback blocks.

3. Democratized and Actionable Learning Paths: Past pathway designs are locked inside premium closed corporate HR setups (e.g., SAP Hub). Skillora AI introduces a modular architectural framework accessible to students, utilizing open learning data vectors to build dynamic, step-by-step career readiness scores and structured milestones.

By fusing semantic embeddings (SBERT) for career role alignment, utilizing structured hybrid keyword-transformer extraction layers, and outputting explicit readiness scores alongside open-source learning milestones, Skillora AI turns isolated analysis observations into clear goals. This literature check confirms that the functional workflow steps of Skillora AI—ranging from multi-format ingestion to customized prioritized learning maps—are backed by sound academic principles and address important unmet needs in modern educational technology.